

# Yi Chen

Ph.D. Candidate in Electrical and Computer Engineering, University of Wisconsin–Madison  
Expected graduation: 2028 | Advisor: Ramya Korlakai Vinayak  
yi.chen@wisc.edu | reid@deepneural.network  
deepneural.network | github.com/kitkatdafa | Google Scholar

## Research Focus

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**Trustworthy AI for retrieval and agents:** Reliable and efficient evaluation of retrieval-augmented generation (RAG) and agentic systems, including factuality verification, conformal risk control under distribution shift, and LLM judges; human feedback, data-centric learning, and pluralistic alignment.

## Education

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### University of Wisconsin–Madison

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|-----------------------------------------------------|------------------------------|
| Ph.D., Electrical and Computer Engineering          | 2023–present (expected 2028) |
| M.S., Electrical and Computer Engineering           | 2022–2025                    |
| B.A., Computer Sciences, <i>Honors in the Major</i> | 2019–2022                    |
| B.A., Mathematics                                   | 2019–2022                    |

## Current Research

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### Conformal Factuality for Retrieval-Augmented Generation (RAG) 2025–2026

- Led a study of conformal factuality across three benchmarks and multiple model families; introduced informativeness-aware metrics and diagnosed vacuity, retrieval shift, and distractor sensitivity.
- Found that lightweight entailment verifiers match or outperform LLM confidence scorers while using under 1% of the FLOPs. Preprint (2026); ICML workshop (2025).

### PALIMPSEST — Efficient Factuality Verification 2026–present

- Proposed a masked-diffusion verifier that masks entity spans and scores length-normalized token recovery; designed reference-grounded and reference-free evaluations against trained checkers and token-probability baselines.
- Designed a matched cloze comparison of Dream-7B and Qwen2.5-7B to isolate reconstruction-based verification behavior; ongoing work extends the approach to conformal factuality under generator shift.

## Industry Experience

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### Meta — Core Ads Growth Engineering 2026

Software Engineer Intern, Machine Learning (Summer 2026)  
Part-time Student Researcher (Fall 2026)

- **Ad Creative Agent (Jun–Jul 2026):** Designed and shipped a production multi-turn, multi-model agent for generating, editing, and iterating on ad creatives from advertiser briefs.
- Architected a ReAct orchestrator with seven tools and four specialized advisors, integrating diffusion-based generation and editing, a learned performance scorer, and a vision-language critic.
- Built a vision-language-model-as-judge evaluation suite with metrics, dashboards, and mock-based tests; developed supervised fine-tuning, model-serving, and live/mock UI infrastructure.

## Publications

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\* denotes equal contribution.

### Preprint

- [1] **Chen, Y.**, Chen, D., Chikodikar, S. M., Yin, C. H., and Vinayak, R. K. "Is Conformal Factuality for RAG-based LLMs Robust? Novel Metrics and Systematic Insights." *arXiv:2603.16817*, 2026.

### Conference Publications

- [2] Tatli, G., **Chen, Y.**, Mason, B., Nowak, R. D., and Vinayak, R. K. "Metric Learning in an RKHS." *Uncertainty in Artificial Intelligence (UAI)*, 2025.
- [3] Vishwakarma, H.\* , **Chen, Y.\***, Namburi, S. S. S. GNVV, Tay, S. J., Vinayak, R. K., and Sala, F. "Re-thinking Confidence Scores and Thresholds in Pseudolabeling-based SSL." *International Conference on Machine Learning (ICML)*, 2025.
- [4] **Chen, Y.** and Vinayak, R. K. "Query Design for Crowdsourced Clustering: Effect of Cognitive Overload and Contextual Bias." *The ACM Web Conference (WWW)*, 2025. **Oral presentation.**
- [5] Chen, D., **Chen, Y.**, Rege, A., Wang, Z., and Vinayak, R. K. "PAL: Pluralistic Alignment Framework for Learning from Heterogeneous Preferences." *International Conference on Learning Representations (ICLR)*, 2025.
- [6] Vishwakarma, H., **Chen, Y.**, Tay, S. J., Namburi, S. S. S., Sala, F., and Vinayak, R. K. "Pearls from Pebbles: Improved Confidence Functions for Auto-labeling." *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [7] Tatli, G., **Chen, Y.**, and Vinayak, R. K. "Learning Populations of Preferences via Pairwise Comparison Queries." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- [8] **Chen, Y.**, Vinayak, R. K., and Hassibi, B. "Crowdsourced Clustering via Active Querying: Practical Algorithm with Theoretical Guarantees." *AAAI Conference on Human Computation and Crowdsourcing (HCOMP)*, 2023.

### Workshop Publications

- [9] **Chen, Y.**, Yin, C. H., Chikodikar, S. M., and Vinayak, R. K. "On the Scoring Functions for RAG-based Conformal Factuality." *ICML Workshop on Reliable and Responsible Foundation Models*, 2025.
- [10] Vishwakarma, H., **Chen, Y.**, Namburi, S. S. S. GNVV, Tay, S. J., Vinayak, R. K., and Sala, F. "PabLO: Improving Semi-Supervised Learning with Pseudolabeling Optimization." *ICML Workshop on Data-Centric Machine Learning Research*, 2024.
- [11] **Chen, Y.** and Vinayak, R. K. "Query Design for Crowdsourced Clustering: Effect of Cognitive Overload and Contextual Bias." *ICML Workshop on Models of Human Feedback for AI Alignment*, 2024.
- [12] Tatli, G., **Chen, Y.**, and Vinayak, R. K. "Learning Populations of Preferences via Pairwise Comparison Queries." *ICML MFPL Workshop*, 2023.
- [13] **Chen, Y.**, Vinayak, R. K., and Hassibi, B. "Crowdsourced Clustering via Active Querying: Practical Algorithm with Theoretical Guarantees." *ICML AI & HCI Workshop*, 2023.
- [14] Tatli, G., **Chen, Y.**, and Vinayak, R. K. "Learning Preference Distributions From Pairwise Comparisons." *International Workshop on Computational Social Choice (COMSOC)*, 2023.

## Selected Research Contributions

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### Reliable Data Annotation and Pseudolabeling

2024–2025

- Co-developed confidence functions and thresholding methods for automated labeling under explicit error constraints; improved coverage by up to 60% over calibration baselines while holding error below 5% (NeurIPS 2024).
- Connected automated-label quality and quantity through a coverage–error objective, leading to an ICML 2025 publication (equal contribution).

### Human Feedback, Preference Learning, and Pluralistic Alignment

2023–2025

- Co-developed PAL, a pluralistic alignment framework that learns shared structure and individual differences from heterogeneous preferences (ICLR 2025).
- Studied identifiability and recovery of preference distributions from limited pairwise comparisons (AISTATS 2024), and nonlinear metric learning in reproducing kernel Hilbert spaces (UAI 2025).
- Designed crowdsourced studies of cognitive overload and contextual bias to understand how question format changes annotation quality; presented the resulting work as an oral at WWW 2025.
- Developed active querying for crowdsourced clustering with theoretical recovery guarantees and evaluation on human-generated data (HCOMP 2023).

## Public Research Artifacts

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**Open-source software:** PAL implementation and datasets (Apache-2.0; ICLR 2025). **Public code and data:** WWW 2025 (Zenodo), AISTATS 2024, and HCOMP 2023.

## Teaching and Mentoring

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### University of Wisconsin–Madison

Teaching Assistant, ECE 539: Introduction to Artificial Neural Networks

Summer 2025, Fall 2025, Fall 2026

- Mentored undergraduate researchers on research projects and scientific writing.

## Professional Service

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**Conference reviewer:** NeurIPS 2026; COLM 2026; UAI 2026; ICLR 2025, 2026; The Web Conference 2025.

**Workshop reviewer:** NeurIPS 2024; ICML 2024.

## Awards and Scholarships

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HCOMP/CI Travel Award, 2023

UW-Madison Conference Presentation Award

UW-Madison Undergraduate Scholarship for Summer Study

James Graaskamp Scholarship

## Technical Skills

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**Research and evaluation:** Agent and LLM-judge evaluation; conformal prediction and factuality; human feedback and preference learning.

**Machine learning:** PyTorch, Hugging Face Transformers, JAX, TensorFlow, scikit-learn; supervised fine-tuning and evaluation.

**Programming:** Python (primary), C/C++, CUDA, Java, JavaScript, Rust, SQL.

**Systems:** Model serving, Docker, Git, AWS (EC2, S3, Lambda), React, Node.js, MongoDB.